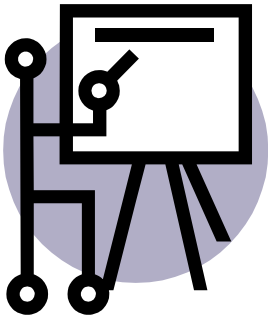
Finite R&D & OpenStudio: Taking the world to another level!

Tech Talk Thursday

Issue 1 4/24/2026



Idk, this looks interesting?

Re-Envisioning the PC Platform

During the late 1980s, personal computers became increasingly popular as manufacturing costs declined and digital technology became more accessible to everyday households.

By the 1990s and early 2000s, personal computing had evolved dramatically. Dedicated graphics cards enabled immersive 3D games such as *Myst* and *Lego Island*, while films like *Toy Story 2* and *Shrek* showcased the growing power of computer-generated animation.

Today, computers no larger than a thumb can perform vastly more operations per second than the Apollo 11 guidance computer that helped land humans on the Moon in 1969.

Now, Finite is re-envisioning the personal computer with a new platform: **Stardust**.

Powered by technology from Finite and OpenStudio, Stardust is designed to be a bold new take on personal computing.

Stardust is our boldest experiment yet — a personal computer platform built for creators, tinkerers, and anyone who still believes computers should be exciting.

Planned software for the platform includes **StarEdit**, **Micron**, **Play**, and **Detail***.

Development of Stardust has only just begun, but expectations for the platform are already high.

More information about Stardust can be found on our website at the end of this newsletter.

* Not all software will be available at launch. Release schedules are subject to delays, testing, and beta availability.

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Lattice Gets a rewrite

Lattice, Our Base implementation runtime for ObjectIR has had it's fair share of issues over the past few months.

Most notably our Virtual machine having it's json reader stripped which caused the codebase to explode with countless issues with the internal parsing.

With this new rewrite, we hope that performance will improve alongside a stable ABI that developers can use with the ObjectIR language. Not only improving developer workflow but also allowing more complex programs to be constructed.

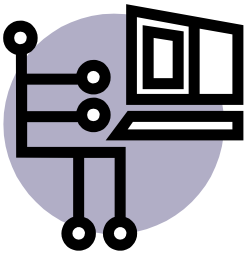
This rewrite should take about a month minimum to get working with our current dev team (1 soul).

Delays with keeping up on other Software will surely add onto more time, but we hope to get a working build out soon!

Tech Research Wizardry:

- ObjectIR Lattice Runtime is Getting a Rewrite: (2)
- We got java "on" C#!
- Construct and Contract see the sun!

Keep reading to see more of our wizardry that we have done since our start in the world.



Ok this is different, is this a person at a pc?

"To catch the reader's attention, place an interesting sentence or quote from the story here."

Standards to the rescue!

During the initial development of ObjectIR back in November, we ran into a major issue: writing complex programs was harder than expected.

Much of this came from our research runtime, Lattice, which was still tied to an outdated ABI that slowed development and made interoperability difficult. At the same time, our native C++ runtime, Iridon, was struggling to keep pace with the evolving ObjectIR specification.

Things began to improve when our lead developer started work on a new project known as MSIR.

MSIR is a tool that reads CoreCLR bytecode such as CIL (Common Intermediate Language) from supported executables and dynamic-link libraries (DLLs), converts the contents into an in-memory data tree, and emits an ObjectIR-compatible output.

This opens the door for a native variant of ObjectIR capable of self-hosting its own runtime and compiler infrastructure, while still integrating with the existing .NET ecosystem.

Alongside development of MSIR, another major gap became clear: ObjectIR lacked a formal standard library for programs running on shared virtual machines.

In this context, a "standard library" follows the same role as projects such as glibc or musl on Linux — providing a common, dependable interface between applications and the underlying runtime.

The new ObjectIR Standard Library specification defines the interfaces runtime developers must implement, enabling portable applications, shared tooling, and a more consistent development experience across all ObjectIR runtimes.

Glorious Generics

Building computer programs in high level languages like C# or java often requires custom classes that implement custom types to allow for more constructive programs to be created.

These are called generics.

And for awhile now, ObjectIR has not had the concept of "Generics", none of that `<T>` nonsense, no `(of Tx)` or `public T Root` type stuff here.

Just plain old inheritance and extends.

With our standard library, we aim to implement Generics support directly into the library with methods to allow runtime developers to construct generic types live for the runtimes to use.

We believe this is the right way to go for ObjectIR tooling right now and will continue integrating great things into the core specification.

Construct your way to victory!

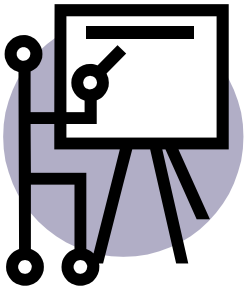
Ever wish your computer would just listen to you when you Tell it to do something?

Look no further to Construct with Intergrated Contract Language support.

No longer shall you be limited to just Object orented language styles, You too can write programs in both Functional and Object Orented languages without the hassle of multiple projects and dependency resolution between them.

You too can write programs that behave how you want, no more weird functions that make no sense, all of your language features provide enough to get you writing amazingly maintainable codebases without the fuss of stuff breaking!

Alongside the functional aspect, Construct Runs on ObjectIR so you also get all of our libraries and the vast tooling that comes with the platform.



Here it is again

Functional saves the day!

Programming can be fun, but what happens when you need to do math.

You write lines of code only to realize there's too much interference with the way your computer runs your "code"*

Look no further to contract!

A functional language designed to allow for developers to write programs that describe their own way they work.

Not only does Contract give you functional aspects but teaming up with Construct, you too can write Object orented programs functionally with better syntax then your powershell install can muster up in a week.

Contract also runs on ObjectIR so you too can use our advanced tooling and backends to allow for developers to target anything and everything that exists!

Why on god does this have to be interesting?

Nobody knows.

Charlie—2026

Java on C#? What!

Ever since ObjectIR arrived mid October, the development team at finite has looked at inspiration on stuff that has happened in the past for guidance for the future, one such project is J#.

J# is a discontinued programming language that was developed to port Java onto the .net platform.

After a few years, Microsoft stopped supporting J# due to rising popularity with C# and the .NET ecosystem.

We brought that back.

Due to research efforts, we are porting Java back to the Dotnet ecosyste, via Java.net which allow's developers to write up-to-date code with up-to-date language features.

All of java at your fingertips!

Finite R&D

This issue was a big doozy to read, but I hope you have had fun reading what's been happening at finite this week

* Not all newsletters are feature complete as Tech-Talk, for more information about us, our projects and more, please check out

https://finite.ovh

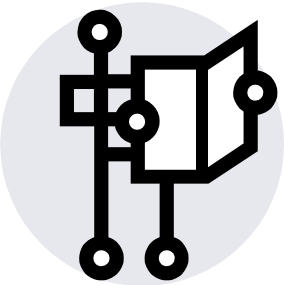
researching the future, one project at a time

Finite R&D is a for profit organization based in Australia designing software for Research to improve the way we use our computers.

Either by having fun making games in games, or Building software development environments that bring the power of the sun into your computer*

Anything you can think of, we look at and see what value we can create of it.

The possibilities are endless



IT'S READING A BOOK!!!
THIS IS INCREDIBLE, This needs to be studied!!!!

Community Creations!

At Finite, we strive to create products / projects that make a change, no matter if it's big or small.

Finite Opens the opportunity to allow for smaller, less known developers or creators to get their name out via the Community section inside our magazine!

Join our discord on our website, navigate to the Community-Creations forum, submit your projects or creations whether tech or not, and the top 10 will be selected for that week to be put in the magazine!